

OXIDATIVE DAMAGE IN RAT TISSUE FOLLOWING EXCESSIVE L-TRYPTOPHAN AND ATHEROGENIC DIETS

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ABSTRACT

Numerous reports were published on the connection between diets containing excessive L-tryptophan and the development of Eosinophilia Myalgia Syndrome. It has been also demonstrated that some cell functions depend on fatty acid composition which can result in increased lipid peroxidation in cells such as macrophages and other inflammatory cells. The purpose of the present study was to investigate the combined effects of an atherogenic diet enriched with tryptophan on lipid peroxidation in rats.

3-week-old CD-1 female rats were fed (3 weeks) control or atherogenic diets and the same diets supplemented with 0.4% or 1.0% L-tryptophan. Liver and skeletal muscle samples from all groups were taken for histology, autoradiography and for determination of lipid peroxidation. Infiltration of cells into fascia of muscle was observed following tryptophan or atherogenic diet consumption. However, no change in 3H-thymidine incorporation into DNA was observed by autoradiography. A significant increase of lipid peroxidation was detected in muscle following consumption of L-tryptophan-rich diets, with no significant difference from control in animals treated with atherogenic diets. Contrastly, a reduced lipid peroxidation was detected in liver of animals treated with excessive tryptophan as well as in animals fed on L-tryptophan and atherogenic diet or atherogenic diet alone.

Our results indicated that excessive dietary tryptophan, when consumed with an atherogenic diet, increased lipid peroxidation in muscle but not in liver. Consumption of these feeding diets with or without supplementation of tryptophan resulted in reduced lipid peroxidation in muscle as well as in liver.

1. INTRODUCTION

Numerous reports concerning diets containing excessive L-tryptophan (Trp) and the development of Eosinophilia Myalgia Syndrome (EMS) were reported (Kaufman *et al.*, 1994). This syndrome which is characterized by scleroderma-like induration, neuromuscular manifestations, arthralgia, pneumonitis and thromboembolic events, has been associated with a preceding history of oral intake of Trp. Even though an epidemic study indicated that a contaminant 1,1'-ethylidenebis (L-tryptophan) (EBT) was involved in EMS (Love *et al.*, 1993), abnormalitis in the metabolism of tryptophan associated with tissue fibrosis and inflammation were also documented in a number of other syndromes such as carcinoid syndrome, scleroderma, rheumatoid arthritis (Freis *et al.*, 1973). Animal studies were controversial but it has been reported that female lewis rats (LEW/N) fed on tryptophan implicated in the EMS epidemic or LEW/N rats fed synthesized EBT, had developed myofascial fibrosis similar to the observed in patients with EMS (Silver *et al.*, 1994).

It has been previously demonstrated that some cell and tissue functions depend on fatty acid composition and that specific types of dietary lipids can alter the function of the immune system by affecting membrane fluidity of macrophages and other cytokine-secreting cells. Such cells are one of the main sources for oxidizing agents in cells and tissues (Robinson *et al.*, 1991; Newsholm *et al.*, 1993).

Dietary tryptophan was previously found to increase platelet aggregation (Mokady *et al.*, 1990), and there was a synergistic effect between atherogenic diet and tryptophan on induction of macrophage uptake of plasma cholesterol (Aviram *et al.*, 1991). Dietary tryptophan supplementation increased plasma lipid peroxidation in control and hypercholesterolemic animal's (Newsholm *et al.*, 1993). Similarly, excessive dietary tryptophan enhanced macrophage cholesterol esterification rate following cell incubation with the plasma obtained from the control and from hypercholesterolemic animals. Excessive dietary tryptophan resulted also in the appearance of numerous TNF- α positive cells indicative of macrophages in muscle and lung of rats fed 3 weeks 0.4%–1% tryptophan (Ronen *et al.*, 1996). Thus, an interaction between tryptophan and dietary lipid which result in induced peroxidation, may also be involved in induction of damage to cell membrane and activation of macrophages and other cytokine producing cells. Another route of interaction between tryptophan and lipids is related to induced kynurenine pathway of tryptophan metabolism. Rudzite *et al.* (1991), also reported on the interaction between lipid metabolism and kynurenine, a metabolite of tryptophan. It has been constantly reported that EMS was associated with infiltration of eosinophils and inflammatory cells in skeletal muscle and was characterized by induced fibrosis in muscle fascia and in lungs.

In order to answer what are the possible environmental factors that elicited such dramatic response to tryptophan enriched diets such as in EMS patients, in the present study the possible interaction between Trp and saturated lipids enriched diets in rats were tested.

2. MATERIALS AND METHODS

3-months-old female rats (Charles River, CD-1) were divided into six groups of 6 animals each, fed for up to 3 weeks on a diet containing 10% protein derived from casein, supplemented with either 8% soybean oil (control) or with 8% coconut oil, 1% cholesterol and 0.5% cholic acid (atherogenic diet) (Anonymous, 1977). These basic diets were further supplemented with 0.4% or 1.0% tryptophan (Sigma, St Louis, MO.). All animals were kept individually in wire cages, maintained at 25°C with light/dark cycles of 12/12 h each. Food and water were supplied ad libitum. Body weights were determined weekly and food intake was calculated on every alternate day. Upon termination of the feeding period, animals were scarified. Liver and muscle samples were taken for the different morphological and biochemical determinations. Gastrocnemius muscles were individually dissected and quickly frozen in isopentane bathed in liquid nitrogen (−180°C) and kept in −70°C until used for the biochemical studies. Other samples of gastrocnemius muscle and liver were fixed in 10% neutral phosphate buffered formaldehyde pH 7.4, dehydrated in graded ethanols, embedded in paraplast (Sherwood medical, USA), sectioned and stained with hematoxylin and eosin (H&E) for general morphology observations. Three animals from each group were injected i.p. with 2μCi/gr body weight of ³H-thymidine (40–50 Cimmole specific activity; Nuclear Research Center, Beer Sheva, Israel). Unstained paraffin section were dipped in NTB II photographic emulsion (Kodak, Rochester N.Y. USA) and kept in light tight boxes (4°C, 3 weeks) for demonstration of radioisotope labeling. Sections were developed (7 min) in D-170 Kodak developer, fixed (10 min) in Na₂S₂SO₃, stained lightly with eosin for light microscope evaluation. Lipid peroxidation was measured in liver and gastrocnemius muscle homogenates using the method of Ernster and Nordenbrand, (1967) The thiobarbituric acid (TBA) reactive substances TBARS assay was used to measure malondialdehyde (MDA). 2,2'-azobisdimethylvaleronitrile (AMVN) was used as the oxidizing agent and the pink reaction color developed was read spectrophotometrically at 532 nm. Data were expressed as nmole MDA per mg tissue. For all measurements statistical analysis was performed using unpaired student's t test and results were reported as mean ± SEM. Analysis of variance (ANOVA) for all groups was performed and p < 0.05 was considered as significance level.

3. RESULTS

In comparison to the normal appearance of gastrocnemius muscle obtained from control muscle (Fig. 1a), a pronounced infiltration of inflammatory cells was observed in the fascia of skeletal muscle from rats that received 1% (Fig. 1b), or 0.4% tryptophan (not shown). These cells appeared to be clustered around perimysial small vessels. Increased cellular infiltration and tissue fibrosis were also observed in tissue sections of gastrocnemius muscle from animals fed atherogenic diets combined with of 1% tryptophan (Fig. 1c). Similar results were obtained with atherogenic diets supplemented with 0.4% tryptophan (not shown).

No such response was observed in gastrocnemius muscle of animals fed tryptophan free atherogenic diet. Autoradiography of ³H-thymidine incorporation into DNA, revealed that only few cells incorporated the label in muscle from control group (Fig. 2a) and that it was not followed by increased proliferation in tissue sections obtained from

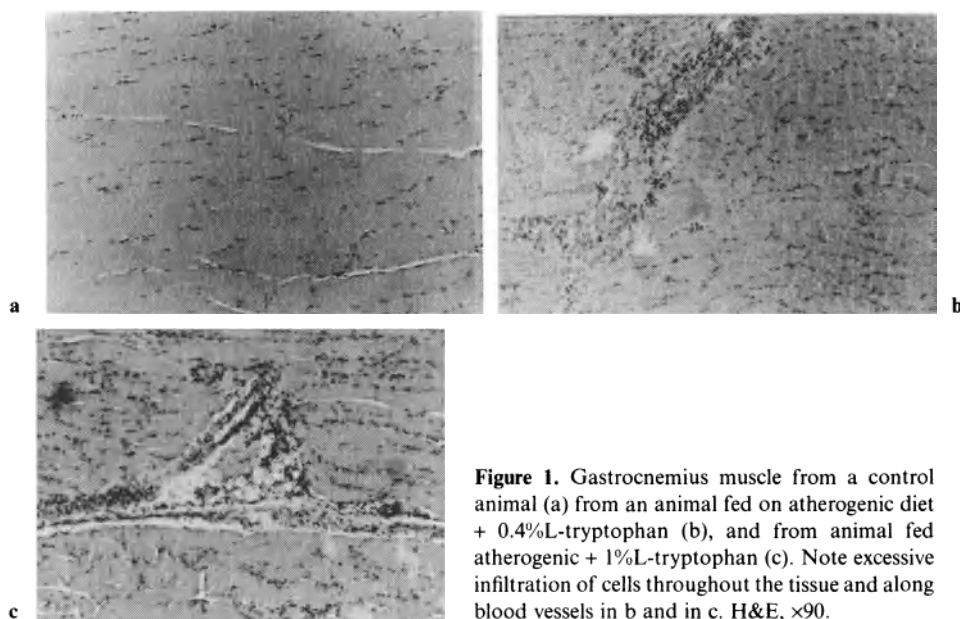


Figure 1. Gastrocnemius muscle from a control animal (a) from an animal fed on atherogenic diet + 0.4%L-tryptophan (b), and from animal fed atherogenic + 1%L-tryptophan (c). Note excessive infiltration of cells throughout the tissue and along blood vessels in b and in c. H&E, $\times 90$.

animals treated either with tryptophan (Fig. 2b) or with combined atherogenic and tryptophan-rich diets (not shown).

Significant ($P < 0.05$) increase of 55%; 54.9% in lipid peroxidation was obtained in gastrocnemius muscle from animals fed a diet containing both 0.4% or 1% tryptophan respectively (Fig. 3).

No similar effect was observed following consumption of atherogenic diet alone or in combinations with tryptophan. Contrastly, the levels of lipid peroxidation appeared to be significantly ($p < 0.05$; $p < 0.01$) reduced (–65%; –50%) in livers from animals fed on the atherogenic diet and the atherogenic diet supplemented with 0.4% or 1%L-tryptophan respectively (Fig. 4).

Morphology of liver tissue from control (Fig. 5a) and from animals treated with atherogenic diets (Fig. 5b) revealed that liver tissue derived from animals fed with the atherogenic diet appeared to contain large lipid droplets in their hepatocytes.

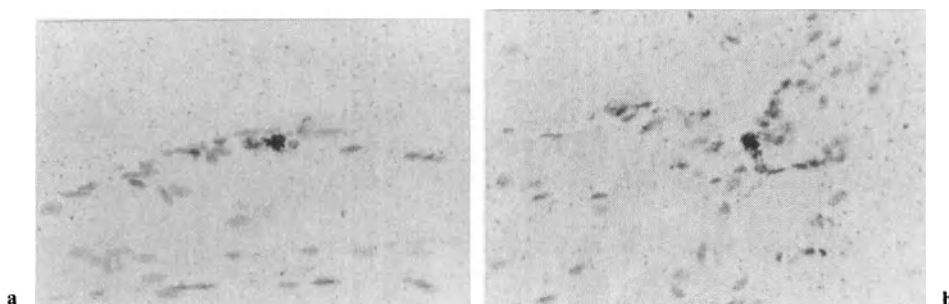


Figure 2. An autoradiogram of muscle following ^3H -thymidine labeling from control (a) and from animal fed atherogenic diet + 1%L-tryptophan (b). $\times 360$.

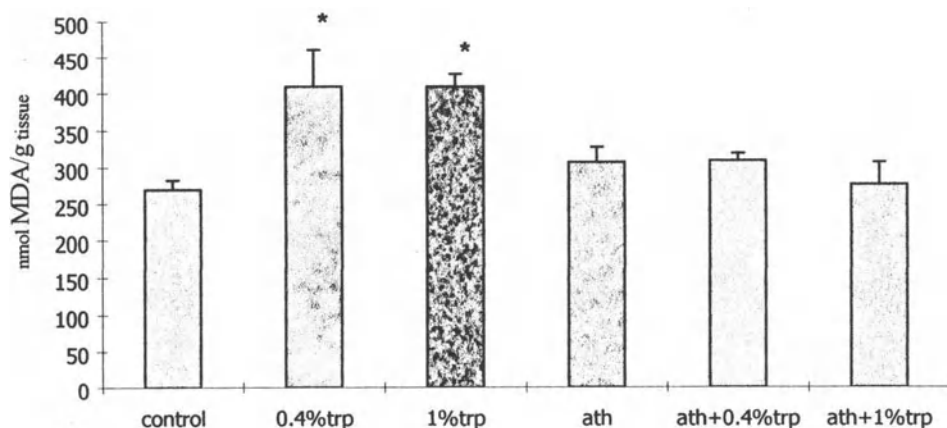


Figure 3. Lipid peroxidation in gastrocnemius muscle from animals fed on control diet, a diet containing 0.4% or 1%L-tryptophan (trp), atherogenic diet (ath) alone or in combinations with L-tryptophan.

4. DISCUSSION

An overall increase of connective tissue and infiltration of cells into muscle were observed in this study. It is known that some cell functions and composition of tissue phospholipid depend also on fatty acid pattern of dietary lipids and that specific types of dietary lipid can alter the function of the immune system. Thus it may be that saturated fatty acids and abnormalities in the metabolism of tryptophan could play a role in the induction of certain inflammatory reactions in muscle.

The purpose of the present study was to test the interactions between saturated fat enriched diet and tryptophan on lipid peroxidation in muscle and liver tissues of rats.

Elevated plasma levels of L-tryptophan were shown to augment the pathway of oxidation to formylkynurenine and synthesis of quinolinic acid (Saito *et al.*, 1993). Excessive levels of tryptophan were also shown to cause a shift to another metabolic pathway, namely the hydroxylation to serotonin. Serotonin, a neurotransmitter and a potent vaso-

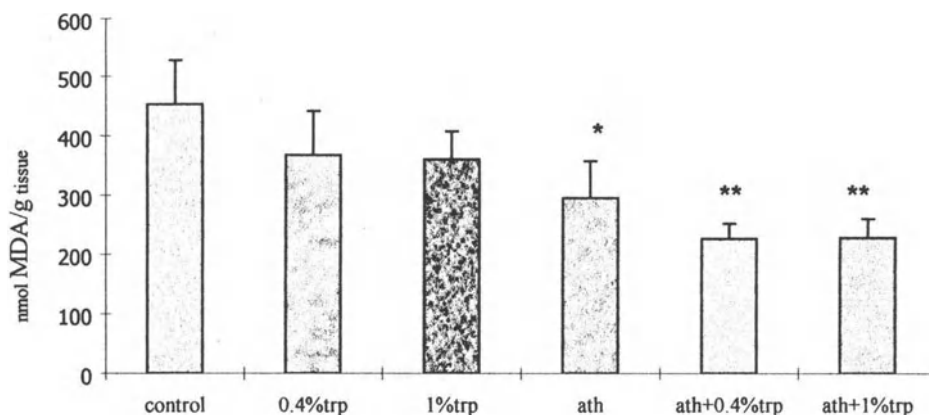


Figure 4. Lipid peroxidation in liver from animals fed on control diet, a diet containing 0.4% or 1%L-tryptophan (trp), atherogenic diet (ath) alone or in combinations with L-tryptophan.

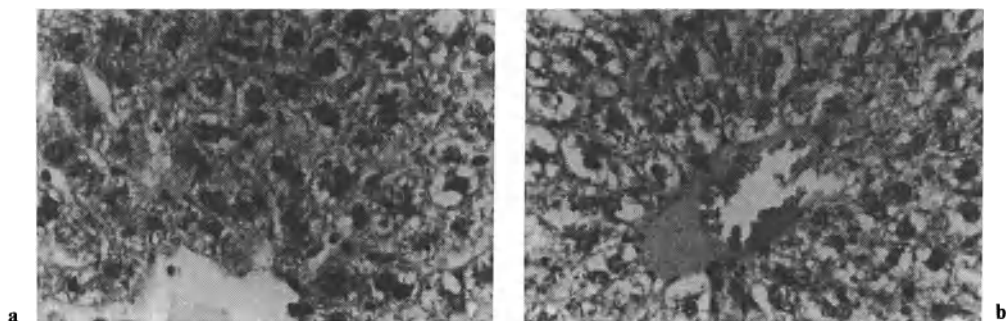


Figure 5. The general appearance of liver tissue from a control animal (a) and from animal fed on atherogenic diet + 1%L-tryptophan (b). Note large lipid droplets in b. H&E, $\times 90$.

constrictor, may also induce myopathy (Schwarcz *et al.*, 1984). Inflammatory cell infiltration into muscle as reported in the present study, might have been due to excessive serotonin production, due to activation of macrophages by the atherogenic diet, or due to increased platelet aggregation. Platelet activation on the other hand, initiates several processes such as smooth muscle cell proliferation, endothelial injury, activation of macrophages and it may also accelerate atherosclerosis (Aviram *et al.*, 1991).

Muscle tissue was constantly reported to be affected by tryptophan treatment in EMS. Similar results were obtained in our study employing CD-1 female rats fed on excessive L-tryptophan. These effects were further augmented by using a combined diet of L-tryptophan and saturated lipids which demonstrated cellular infiltration and tissue fibrosis in gastrocnemius muscle (Gross *et al.*, 1996). As reported previously (Ronen *et al.*, 1996) some of the infiltrated cells were shown to be TNF- α positive, a feature indicative of the presence of macrophages. Activated macrophages are known to produce $O_2^{\cdot -}$. Infiltration of large numbers of macrophages into a localized site, followed by their activation, was shown to produce an intense localized oxidative stress (Reznick *et al.*, 1993). In addition, enzymatic cleavage of the indole ring of tryptophan by indoleamine dioxygenase (IDO) and tryptophan dioxygenase (TDO), produced $O_2^{\cdot -}$ (Halliwell *et al.*, 1985; Christen *et al.*, 1990). It is thus possible that one of the tryptophan metabolites, kynurenine or quinolinic acid that are formed by IDO and TDO are also involved in inducing this response. IDO cleaves tryptophan in all tissues except in liver, where TDO is present. Depending on TDO enzyme activity, the rate of tryptophan cleavage in liver is known to be lower in comparison to the rate of tryptophan cleavage by IDO in all the other tissues (Saito *et al.*, 1993). This can explain the different in lipid peroxidation extent in liver compared to muscle as observed in the present study, resulting in relatively reduced amounts of $O_2^{\cdot -}$ generated in liver in comparison to muscle and other tissues. The extent of lipid peroxidation in the liver reflects the relative amounts of unsaturated fatty acids. In the case of atherogenic diet, which contained saturated fatty acids, lipid peroxidation was reduced as shown in the present study. Liver is one of the organs with the highest antioxidant enzyme activities in the body and has detoxification mechanism as one of its major functions (Waston *et al.*, 1994; Parkhouse *et al.*, 1995). Malondialdehyde is a biomarker of lipid peroxidation, which accumulates in the tissue and is used as an indication of lipid peroxidation stress. Increased lipid peroxidation, which is a marker for free radicals production, may indicate that tryptophan induces tissue damage, as seen in muscle (Fig. 5). The results obtained in the present study indicated that lipid

peroxidation in the liver was reduced in all the groups fed on atherogenic diet containing high levels of saturated fatty acids. It may thus appear that liver has a special protection mechanisms against lipid peroxidation probably due to excessive formation of certain tryptophan metabolites which are known to be antioxidants or because the saturated diets are less prone to oxidation (Cristen *et al.*, 1990).

At normal rates of generation, some free radicals are useful in the human body. When free-radicals generation exceeds the capacity of the antioxidant defense systems, the result is the occurrence of oxidative stress. Production of free radicals by phagocytes is useful in killing foreign organisms, but it also can harm the phagocyte itself and the surrounding tissues. In some human diseases, the harm outweighs these beneficial effects.

One of the targets of free radical attack in human cells is the lipids of the membranes. Free poly unsaturated fatty acid (PUFAs) and other FA which are incorporated into the membrane phospholipid are readily attacked by free radicals, thereby becoming oxidized to lipid peroxides (Reznick *et al.*, 1993). Lipid peroxidation of cell membranes results in decreased membrane fluidity, inability to maintain ionic gradients, cellular swelling and tissue inflammation (Parkhouse *et al.*, 1995). Lipid peroxidation, which is indicative of free radical attack in the tissue and a marker for toxic and tissue damage, had increased in muscle from animals treated with excessive tryptophan. Among the various ways for an increase of free radical production, is the increase of O_2^- generation in the mitochondrial electron transport chain, or increase of metal catalyzed free radical production due to mechanical and morphological damage to muscle (Waston *et al.*, 1994). The hydroxyperoxides produced in this process can induce stimulation of cell proliferation and cause induction of cytokine secretion (Esterbauer *et al.*, 1993) as shown in EMS. Our study demonstrated that excessive dietary tryptophan, when consumed with saturated fatty acids and cholesterol, might mediate inflammation, tissue damage and fibrosis in skeletal muscle as seen in EMS. The combination of the atherogenic diet and tryptophan however, decreased lipid peroxidation probably because high levels of saturated fatty acids incorporated into muscle membrane. In normal fed animals, the fraction of total plasma tryptophan bound to albumin is about 85–90%. This equilibrium can be shifted under conditions which raise plasma non-esterified fatty acid (NEFA) concentrations, such as during fasting or stress (Peter *et al.*, 1991). This is because NEFAs compete with tryptophan for binding sites on the albumin molecule, therefore, when the concentration of NEFAs rise, tryptophan is displaced from the albumin molecule, raising the concentration of free tryptophan. It thus appears that in combination of tryptophan and the fatty acids diet, the actual concentrations of unbound tryptophan in the plasma could be much higher than the calculated values. The liver is the most important organ influencing plasma tryptophan concentration. There is no storage pool in the body for amino acids that are ingested in excess for immediate needs.

Therefore, once the body's demand for the amino acid has been met, the surplus tryptophan is oxidized to provide energy or is converted to fat. The extent of removal of tryptophan by the liver following a meal is influenced by the amount and activity of tryptophan-2,3-dioxygenase, the principal enzyme regulating tryptophan oxidation. The enzyme activity is controlled by its substrate, the tryptophan. In control diet containing unsaturated fatty acids derived from soybean oil and supplemented with tryptophan the effect was augmented as expressed by high levels of lipid peroxidation. The supply of tryptophan to the liver is determined by its uptake from the diet and then by extent of which tryptophan is bound to serum albumin. However, the ability of amino acid catabolic enzyme to adjust to amino acid intake is limited (Peter *et al.*, 1991).

In our study however, it was found that although hepatocytes from animals fed on tryptophan and atherogenic diet were rich in lipid droplets, the levels of lipid peroxidation were not elevated in comparison to the control diets. Since polyunsaturated fatty acids (PUFAs) were also shown to suppress the activation of the immune system (Peter *et al.*, 1991), additions of saturated fatty acids elevated the unbound tryptophan in the plasma and activate the immune system. On the other hand, increasing the relative amounts of saturated fatty acid as in atherogenic diet, reduces the total unsaturated acid pool thus acting in protection against elevated lipid peroxidation as observed here on muscle as well as in liver.

Our study demonstrated is that there exists an interaction between tryptophan and dietary lipids in relation to lipid peroxidation in muscle but not in liver. Our results contribute to the understanding of how dietary fatty acids in combination with tryptophan could affect target tissues and result in damaging or protective responses in these tissues.

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